MIL-C-46237(Ord)
24 June 1960

# MILITARY SPECIFICATION
# CARTRIDGE, AIRCRAFT EJECTION SEAT CATAPULT, M30A1

## 1. SCOPE

1.1 This specification covers the Cartridge, Aircraft Ejection Seat Catapult, M30A1 for use with cartridge actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | | |
|---|---|---|
| MIL-G-2550 | - | General Specification for Ammunition, Except Small Arms Ammunition |
| MIL-E-5272 | - | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |

STANDARDS

Military

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - | Inspection Terms and Definitions |
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-639 | - | Surface Roughness Values Preferred for Army Ordnance Materiel |

FSC 1375

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DRAWINGS

Ordnance Corps

| | | |
|---|---|---|
| D-8593630 | - | Assembly - Cartridge, Aircraft Ejection Seat Catapult, M30A1 |
| F-8797637 | - | Ammo Container Metal, M21 (Modified) for Cartridge, Aircraft Ejection Seat Catapult, M30A1 |
| F-8797638 | - | Box, Packing, Ammunition for Cartridge, Aircraft Ejection Seat Catapult, M30A1, in Metal Container, M21 (Modified) |
| IEL-8593630 | - | Inspection Equipment List for Cartridge Aircraft Ejection Seat Catapult, M30A1 |

PUBLICATIONS

Frankford Arsenal Manual for Functional Tests - Cartridge Actuated Devices for Aircraft Ejection Systems

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION - 49 CFR 71-78

Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles. (The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78".)

3. REQUIREMENTS

3.1 General.- Materials, parts, assemblies, chemical and physical requirements shall conform to the applicable drawings, referenced specifications and the following:

3.2 Preproduction Sample.- Unless otherwise directed by the contracting officer, a preproduction sample shall be required.

3.3 Protective Finish.- When protective finish is required by component details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.



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3.4 Primer.- The primer shall conform to the applicable drawing and referenced specification. Prior to assembly into a cartridge the primer shall be visually examined and x-rayed.

3.5 Cartridge Assembly.- Components shall be combined to produce an assembly which conforms to the requirements of Drawing D-8593630.

3.5.1 Vibration.- The assembled cartridge shall withstand vibration as prescribed in Specification MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.6 Functional Requirements.

3.6.1 The cartridge shall not fail to fire.

3.6.2 Powder Charges.- The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements (see 6.1).

3.6.3 Function.- The cartridge when assembled in an M2 Catapult shall propel a 300 pound mass to comply with the requirements of Table I.

TABLE I

| Temperature °(Degrees) F(Fahrenheit) | Minimum Velocity Ft/Sec | Max. Acceleration (g's) | | Max. Rate Of Change Of Acceleration G/Sec | Ignition Delay Milliseconds |
|---|---|---|---|---|---|
| | | Load Cell | Pressure Time | | |
| 65° to 75°F | 38 | 12 | 14 | 150 | 150 |

3.7 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, specifications and the following:

3.7.1 Metal Defects.- All metal parts shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.7.2 Surface Finish.- The requirements for surface finish are as detailed by the applicable drawing and referenced specification. The Ordnance Standard Finish Blocks prescribed in Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

3.7.3 Burr.- No cartridge shall have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the cartridge.

3.7.4 Foreign Matter.- No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions.- The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 to define the inspection terms used. Inspection shall be in accordance with Specification MIL-G-2550 except as specified herein.

4.1.1 Contractor Inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor Quality Assurance System.- The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification and other applicable documents referenced in this specification. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.3 Government Verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction Inspection.- Prior to the commencement of regular production, the contractor shall submit a preproduction sample to a designated Government approved facility for evaluation. The sample shall consist of 30 cartridges and 10 sets of unassembled components including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All



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4.2 (cont.)

parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the preproduction sample shall be inspected for all requirements of the drawings and specifications including all those characteristics listed in 4.3.3 and the tests in 4.3.4. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions.- Alternative inspection procedures and inspection equipment may be used by the contractor when such procedures and equipment provide, as a minimum, quality assurance required in the contractual documents. The inspection procedures and equipment specified in this specification will prevail in case of dispute.

4.3.1 Material.- Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification of the material and part, the applicable requirements, the specific tests performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction sample. Prior to inspection, the inspector shall determine from the information stated on the test reports whether the material submitted complies with the requirements. Cartridges made from material which does not comply shall be rejected.

4.3.2 Lot.

4.3.2.1 Submittal of Product.- The contractor shall submit the product in accordance with Standard MIL-STD-105.

4.3.2.2 Size of Lot.- Unless otherwise specified by the procuring agency an ammunition lot shall consist of 600 cartridges, maximum, exclusive of test samples, and shall consist of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Igniter from not more than one lot.
d. Primers from not more than one lot.

4.3.3 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter.



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TABLE II

| Lot Size | Sample Size | Acceptance Numbers Major | Minor |
|---|---|---|---|
| 1 - 300 | 100% Examination | 0 | 0 |
| 301 -1300 | 110 | 1 | 2 |
| 1301 -3200 | 150 | 2 | 3 |

TABLE III

| No. of Defects in Class | Total Major Defects Allowable |
|---|---|
| 2 - 3 | 1 |
| 4 - 7 | 2 |
| 8 -19 | 3 |

| No. of Defects in Class | Total Minor Defects Allowable |
|---|---|
| 2 - 3 | 2 |
| 4 - 7 | 3 |
| 8 -16 | 4 |

4.3.3.1 Classification of Defects.

4.3.3.1.1 Cartridge, Aircraft Ejection Seat Catapult, M30A1, Drawing D-8593630.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| a. Any part missing | Visual |
| b. Improper assembly | Visual |
| c. Mixed ammunition | Visual |
| d. Metal defective | Visual |
| e. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 1. Cartridge profile failure, requiring more than 20 pounds dead weight to insert in concentricity receiver gage | Gage |
| 2. Diameter of head, max. | Gage |
| 3. Diameter of body, max. | Gage |
| 4. Length to datum diameter | Gage |



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4.3.3.1.1 (cont.)

| Defect | Method of Inspection |
|---|---|
| MAJOR (cont.) | |
| 5. Dent, failure to chamber in concentricity receiver gage with 20 pounds dead weight | Gage/Visual |
| 6. Foreign matter | Visual |
| 7. "O" ring split, cracked or cut | Visual |
| 8. Burr | Visual |
| MINOR | |
| 41. Total length | Gage |
| 42. Dent, which will chamber in concentricity receiver gage with 20 pounds dead weight, max. | Gage/Visual |
| 43. Draw scratch | Visual |
| 44. Fold, wrinkle, buckle or bulge | Visual |
| 45. Varnish inadequate or missing | Visual |
| 46. Surface finish improper | Visual |

4.3.3.1.2 Ammunition Container Metal, M21 (Modified) for Cartridge, Aircraft Ejection Seat Catapult, M30A1, Drawing F-8797637.

| | |
|---|---|
| MAJOR | |
| None | |
| MINOR | |
| 41. Air test failure | Approved Test |
| 42. Quantity of contents incorrect | Visual |
| 43. Content loose | Manual/Visual |
| 44. Separator, filler or liner missing | Visual |
| 45. Marking, missing, incorrect or illegible | Visual |

4.3.3.1.3 Box, Packing, for Cartridge, Aircraft Ejection Seat Catapult, M30A1, in Metal Container, Drawing F-8797638.

| | |
|---|---|
| MAJOR | |
| None | |
| MINOR | |
| 41. Contents loose | Manual/Visual |
| 42. Hardware missing, loose, broken or improperly assembled | Visual |



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4.3.3.1.3 (cont.)

| Defect | Method of Inspection |
|---|---|
| MINOR (cont.) | |
| 43. Strapping missing, loose, broken or improperly Assembled | Visual |
| 44. Contents of box exposed | Visual |
| 45. Box damaged | Visual |
| 46. Marking missing, incorrect or illegible | Visual |
| 47. Number of containers incorrect | Visual |
| 48. Car seal missing, unsealed or improperly positioned | Visual |

4.3.3.2 Primer Examination.- Prior to assembly into a cartridge, each primer shall be visually examined for any discoloration of the foiling paper or a malformation of any kind. Any discoloration of the foiling paper or malformation or failure to comply with the requirements of the drawing and referenced specification shall be cause for rejection of the primer.

4.3.4 Inspection Testing.- The following tests shall be performed in accordance with the provisions of 4.4, except as provided in 4.3.

4.3.4.1 Protective Finish.- The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.4.2 Primer X-Ray Inspection.- Each primer shall be x-rayed. Failure of any primer to comply with the applicable requirements of 3.4 shall be cause for rejection of the primer.

4.3.4.3 Cartridge X-Ray Inspection.- Each cartridge shall be x-rayed. Failure of any cartridge to conform with the drawing requirements shall be cause for rejection of the cartridge.

4.3.4.4 Airtightness.- Each cartridge shall be subjected to the minimum pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.4.5 Vibration Test, Preproduction Sample.- The thirty assembled, x-rayed cartridges of the preproduction sample shall be subjected to vibration as directed in 3.5.1 and again x-rayed as directed in 4.4.2 for evidence of any disintegration of the primer or propellant or igniter or derangement of any components. Evidence of any disintegration or derangement shall be cuase for rejection of the lot.



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4.3.4.5.1 Vibration Test, Production Sample.- When it becomes necessary to make a change in the lot of propellant or igniter or primers after production is initiated, fifteen assembled cartridges from the first production lot thereafter shall be tested in accordance with 4.3.4.5.

4.3.4.6 Functioning.- The thirty assembled cartridges from each preproduction sample after the vibration test and x-ray, and the fifteen assembled cartridges from each subsequent lot sample shall be temperature conditioned and functionally tested. Failure of any cartridge to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.5 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL-8593630, except as provided in 4.3. The Government will determine by means of its monitoring program, that the contractor has available and correctly utilizes gaging, measuring and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test Methods and Procedures.

4.4.1 Primer X-Ray.- Prior to assembly into a cartridge, the primer shall be x-rayed with the side of the cup perpendicular to the rays to determine compliance with 3.4. X-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.4.2 Cartridge X-Ray.- The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of an x-ray machine approved by the contracting officer. The x-ray negative, properly identified, shall be forwarded to the procuring agency.

4.4.3 Airtightness.- The method for determining airtightness shall be made by use of the equipment listed on the inspection equipment list except as provided in 4.3.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with Specification MIL-E-5272, Procedure XII.

4.4.5 Functional Tests.- These tests will be performed by the Government:

4.4.5.1 Preproduction Sample.- Cartridges of the preproduction sample shall be temperature conditioned in accordance with 4.4.5.3 as follows: thirty cartridges at plus 65 degrees Fahrenheit (F) to plus 75 degrees (F) and then tested in accordance with 4.4.5.4.



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4.4.5.2 Production Lot Sample.- Cartridges of the production lot sample shall be temperature conditioned in accordance with 4.4.5.3 as follows: fifteen cartridges at plus 65 degrees Fahrenheit (F) to plus 75 degrees (F) and then tested in accordance with 4.4.5.4.

4.4.5.3 Temperature Conditioning.- The cartridges shall be loaded into acceptable M2 Catapults with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test catapult device similar to the catapults undergoing tests.

4.4.5.4 Test Procedures.- Upon completion of the required temperature conditioning the sample shall be ballistically tested in accordance with Frankford Arsenal Manual of Functional Tests for Cartridge Actuated Devices for Aircraft Ejection Systems within five minutes after removal from the conditioning chamber.

4.4.5.5 Recordings.- The velocity, ignition delay, acceleration and rate of change of acceleration of the propelled mass shall be recorded and forwarded to the responsible procuring agency.

4.4.5.6 Retest.- Defective Test Equipment.- When it has been determined that the occurrence of failures in the functional tests is directly attributable to defective test equipment, an additional sample shall be provided and tested in accordance with 4.3.4 and 4.4.

## 5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Packing and marking shall be in accordance with Drawings F-8797637 and F-8797638 and Standard MIL-STD-129.

5.2 Delivery.- Delivery shall be made in accordance with Interstate Commerce Commission Rules and Regulations listed in 2.2.

5.3 Data Cards.- Data card information shall be in accordance with Specification MIL-G-2550.

## 6. NOTES

6.1 Advisory loading data:

| | |
|---|---|
| Black Powder, Grade A-1 | 50.0 grains minus 2.0 grains |
| Propellant, Type M-6 | 1270.0 grains minus 15.0 grains |

Propellant Grain Dimensions:

| | |
|---|---|
| Length | 1.15 inches |
| Diameter | .631 inches |
| Perforation Diameter | .063 inches |
| Web | .110 inches |



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6.2 Supplementary Quality Assurance Provisions.- For the procurement of components (except explosive components) inspection shall be in accordance with the applicable drawing, supplementary quality assurance provisions (SQAPs) and Specification MIL-C-45407 (Ord), Components (Except Explosive Components) of Cartridges for Cartridge Actuated Devices. Applicable SQAPs are listed on List of Supplementary Quality Assurance Provisions (LSQAP) 8593630. Copies of the aforementioned specification, drawings, LSQAP and SQAPs may be obtained from the Industrial Group, U.S. Army Ordnance Arsenal, Frankford, Philadelphia 37, Pa.

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